

## AlCu4Mg1; AA2024-T6

| Component | Wt. %       | Component    | Wt. %     | Component | Wt. %    |
|-----------|-------------|--------------|-----------|-----------|----------|
| Al        | 90.7 - 94.7 | Mg           | 1.2 - 1.8 | Si        | Max 0.5  |
| Cr        | Max 0.1     | Mn           | 0.3 - 0.9 | Ti        | Max 0.15 |
| Cu        | 3.8 - 4.9   | Other, each  | Max 0.05  | Zn        | Max 0.25 |
| Fe        | Max 0.5     | Other, total | Max 0.15  |           |          |

### Material Notes:

Weldability = C; Stress Corrosion Cracking Resistance = B; General Corrosion Resistance = D (A = best; E = worst). Good machinability and surface finish capabilities. A high strength material of adequate workability. Has largely superceded 2017 for structural applications.

**Uses:** Aircraft fittings, gears and shafts, bolts, clock parts, computer parts, couplings, fuse parts, hydraulic valve bodies, missile parts, munitions, nuts, pistons, rectifier parts, worm gears, fastening devices, veterinary and orthopedic equipment, structures.

Some data provided by Alcoa.

Data points with the AA note have been provided by the Aluminum Association, Inc. and are NOT FOR DESIGN.

| Physical Properties          | Metric           | English                            | Comments                                                                                                                                                 |
|------------------------------|------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Density                      | 2.78 g/cc        | 0.1 lb/in <sup>3</sup>             | AA; Typical                                                                                                                                              |
| <b>Mechanical Properties</b> |                  |                                    |                                                                                                                                                          |
| Hardness, Brinell            | 125              | 125                                | 500 kg load/10 mm ball                                                                                                                                   |
| Hardness, Knoop              | 157              | 157                                | Estimated from Brinell                                                                                                                                   |
| Hardness, Rockwell A         | 48               | 48                                 | Estimated from Brinell                                                                                                                                   |
| Hardness, Rockwell B         | 78               | 78                                 | Estimated from Brinell                                                                                                                                   |
| Hardness, Vickers            | 142              | 142                                | Estimated from Brinell                                                                                                                                   |
| Tensile Strength, Ultimate   | Min 427 MPa      | Min 61900 psi                      |                                                                                                                                                          |
| Tensile Strength, Yield      | Min 345 MPa      | Min 50000 psi                      |                                                                                                                                                          |
| Elongation at Break          | 5 %              | 5 %                                |                                                                                                                                                          |
| Modulus of Elasticity        | 72.4 GPa         | 10500 ksi                          | Estimated from other heat treatments                                                                                                                     |
| Poisson's Ratio              | 0.33             | 0.33                               | Estimated from other heat treatments                                                                                                                     |
| Fatigue Strength             | 124 MPa          | 18000 psi                          | 500,000,000 cycles, completely reversed, R. R. Moore Machine and specimen                                                                                |
| Machinability                | 70 %             | 70 %                               | 0-100 Scale (A=90, B=70, C=50, D=30, E=10)                                                                                                               |
| Shear Modulus                | 27 GPa           | 3920 ksi                           | Estimated from similar Al alloys.                                                                                                                        |
| Shear Strength               | 283 MPa          | 41000 psi                          |                                                                                                                                                          |
| <b>Electrical Properties</b> |                  |                                    |                                                                                                                                                          |
| Electrical Resistivity       | 4.49e-006 ohm-cm | 4.49e-006 ohm-cm                   | AA; Typical at 68°F                                                                                                                                      |
| <b>Thermal Properties</b>    |                  |                                    |                                                                                                                                                          |
| CTE, linear 68°F             | 23.2 µm/m-°C     | 12.9 µin/in-°F                     | AA; Typical; Average over 68-212°F range.                                                                                                                |
| CTE, linear 250°C            | 24.7 µm/m-°C     | 13.7 µin/in-°F                     | Average over the range 20-300°C                                                                                                                          |
| Specific Heat Capacity       | 0.875 J/g-°C     | 0.209 BTU/lb-°F                    |                                                                                                                                                          |
| Thermal Conductivity         | 151 W/m-K        | 1050 BTU-in/hr-ft <sup>2</sup> -°F | AA; Typical at 77°F                                                                                                                                      |
| Melting Point                | 502 - 638 °C     | 935 - 1180 °F                      | AA; Typical range based on typical composition for wrought products 1/4 inch thickness or greater. Eutectic melting is not eliminated by homogenization. |
| Solidus                      | 502 °C           | 935 °F                             | AA; Typical                                                                                                                                              |
| Liquidus                     | 638 °C           | 1180 °F                            | AA; Typical                                                                                                                                              |
| <b>Processing Properties</b> |                  |                                    |                                                                                                                                                          |
| Annealing Temperature        | 413 °C           | 775 °F                             |                                                                                                                                                          |
| Solution Temperature         | 256 °C           | 493 °F                             |                                                                                                                                                          |
| Aging Temperature            | 191 °C           | 375 °F                             | 8 to 16 hr at temperature                                                                                                                                |

### References for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's disclaimer and terms of use regarding this information.